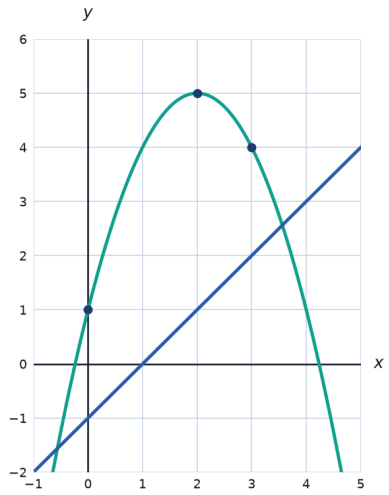
Completing the square (Extended)

Core Concepts: Completing the square (Extended)

1 For the model $f(x) = 2x + 6$ in completing the square (extended), evaluate $f(2)$.



a. 10

b. 12

c. 4

d. 14

2 Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ and $x = 5$

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

3 A data set recorded for completing the square (extended) consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 2x + 6$ in completing the square (extended), evaluate $f(3)$.

a. 12

b. 14

c. 6

d. 16

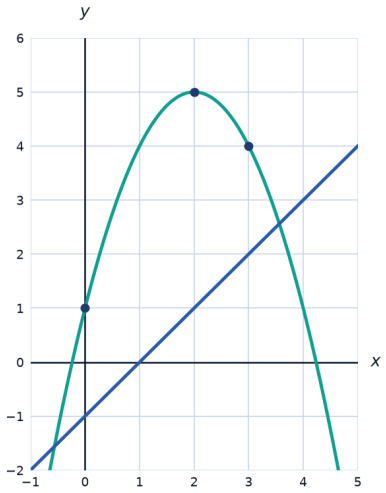
5 Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$



6 A data set recorded for completing the square (extended) consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

Graphical & Geometric Interpretation

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7 For the model $f(x) = 2x + 6$ in completing the square (extended), evaluate $f(4)$.

a. 14

b. 16

c. 8

d. 18

8 Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ and $x = 7$

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

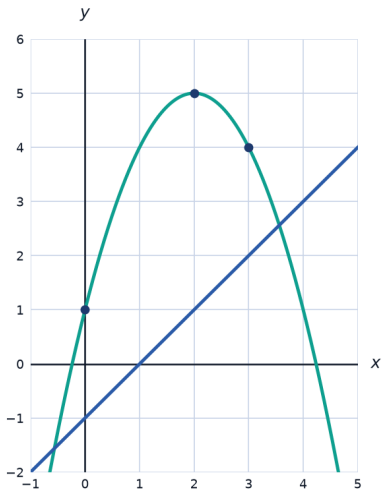
9 A data set recorded for completing the square (extended) consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

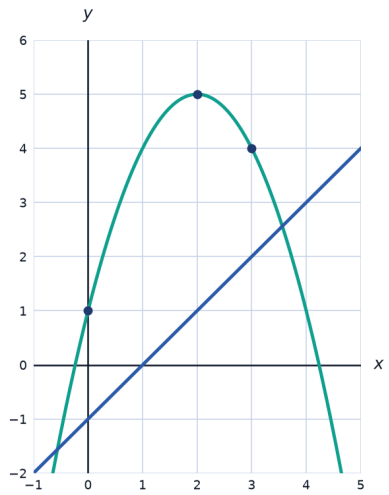
d. 24.5



Exam-Style Problem Solving & Modeling

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10 For the model $f(x) = 2x + 6$ in completing the square (extended), evaluate $f(5)$.



a. 16

b. 18

c. 10

d. 20

11 Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ and $x = 8$

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

12 A data set recorded for completing the square (extended) consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

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